

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q24-2451

Prairie 138 kV
August 24, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2451. The Interconnection Customer proposes a 400.0 MW solar photovoltaic facility interconnecting at the Prairie 138 kV substation in the Virginia area. The request was received on September 16, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 468 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$45,608,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: PRAIRIE – TODDS TAVERN DP 138 KV
Contingency: PRAIRIE – NORTH ANNA 138 KV
Loading: 170.0% (Rating: 224 MVA)
DFAX: 100.0%
Pre-project loading: 8.6%
2. Facility: N ANNA – PRAIRIE 138 KV
Contingency: PRAIRIE – TODDS TAVERN DP 138 KV
Loading: 157.7% (Rating: 241 MVA)
DFAX: 100.0%
Pre-project loading: 8.0%
3. Facility: NI RIVER DP – CHANCELLOR 138 KV
Contingency: N ANNA 345-138 kV TX
Loading: 149.7% (Rating: 355 MVA)
DFAX: 72.1%
Pre-project loading: 68.5%
4. Facility: NORTH ANNA 345/138 kV Transformer Bank
Contingency: STORY CITY – CHANCELLOR 138 KV line
Loading: 143.9% (Rating: 505 MVA)
DFAX: 88.4%
Pre-project loading: 73.9%
5. Facility: GULFGAE – WAKE 138 KV
Contingency: N ANNA 345-138 kV Auto
Loading: 134.1% (Rating: 223 MVA)
DFAX: 34.1%
Pre-project loading: 72.7%

6. Facility: BORAH – TODDS TAVERN DP 138 KV L
Contingency: ACEC QUINCY – NI RIVERDP 138 KV
Loading: 133.4% (Rating: 203 MVA)
DFAX: 30.7%
Pre-project loading: 73.0%
7. Facility: CHANCELLOR – STORY CITY SUBSTATION 138 KV
Contingency: N ANNA 345/138kV Transformer Bank
Loading: 128.4% (Rating: 522 MVA)
DFAX: 72.1%
Pre-project loading: 73.2%
8. Facility: STORY CITY – GULFGATE SUBSTATION 138 KV
Contingency: N ANNA 345-138 kV XFMR
Loading: 127.9% (Rating: 273 MVA)
DFAX: 34.1%
Pre-project loading: 77.8%
9. Facility: LADYSMITH CT – STORY CITY 138 KV
Contingency: N ANNA 345 kV/138 kV XFMR
Loading: 127.8% (Rating: 213 MVA)
DFAX: 38.0%
Pre-project loading: 56.5%
10. Facility: GARRISONVILLE – WAKE SUBSTATION 138 KV
Contingency: NORTH ANA 345 kV/138 kV Auto
Loading: 102.6% (Rating: 173 MVA)
DFAX: 25.4%
Pre-project loading: 44.0%
11. Facility: BORAH – NORTH ANNA 138 KV
Contingency: STORY CITY – CHANCELLOR 138 KV Line
Loading: 100.9% (Rating: 309 MVA)
DFAX: 27.3%
Pre-project loading: 65.6%
12. Facility: BORAH – NI RIV DP 138 KV
Contingency: CHANCELLOR – STORY CITY 138 KV Line
Loading: 100.1% (Rating: 245 MVA)
DFAX: 7.1%
Pre-project loading: 88.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
TODDS TAVERN DP SUBSTATION – PRAIRIE 138 KV Line	Rebuild line with higher-capacity conductor	\$7,958,000	100% (\$7,958,000)
N ANNA – PRAIRIE 138 KV	Rebuild line with higher-capacity conductor	\$13,733,000	100% (\$13,733,000)
NI RIVER DP SUBSTATION – CHANCELLOR 138 KV	Rebuild line with higher-capacity conductor	\$3,185,000	46% (\$1,467,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
NORTH ANNA 345/138 kV Transformer Bank	Replace transformer with higher-rated unit	\$15,453,000	41% (\$6,323,000)
GULFGATE – WAKE 138 KV	Rebuild line with higher-capacity conductor	\$7,838,000	38% (\$2,987,000)
BORAH – TODDS TAVERN DP 138 KV L	Rebuild line with higher-capacity conductor	\$10,879,000	34% (\$3,677,000)
CHANCELLOR – STORY CITY SUBSTATION 138 KV	Rebuild line with higher-capacity conductor	\$2,987,000	38% (\$1,139,000)
STORY CITY – GULFGATE SUBSTATION 138 KV	Rebuild line with higher-capacity conductor	\$3,220,000	38% (\$1,227,000)
LADYSMITH CT – STORY CITY 138 KV	Rebuild line with higher-capacity conductor	\$15,994,000	38% (\$6,099,000)
WAKE – GARRISONVILLE 138 KV	Reconductor line	\$839,000	38% (\$319,000)
BORAH – NORTH ANNA 138 KV line	Reconductor line	\$1,178,000	22% (\$263,000)
BORAH – NI RIV DP 138 KV	Reconductor line	\$5,024,000	8% (\$416,000)
	Total allocated to project		\$45,608,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.